

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 03-04-2025	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
March 2025



Contents

1	NIGHT1: Contingency tables	2
2	DAY1: Contingency tables	8
3	NIGHT2: Contingency tables	11
4	DAY2: Contingency tables	16
5	NIGHT3: Contingency tables	18
6	DAY3: Contingency tables	23

1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4214	1586	203
	4.2 < ws < 7.6	2622	3481	1064
	ws > 7.6	336	1583	4777
Number of points 19866 ; PC=62.8%; EBD=2.7%; POD1=58.8%; POD2=52.3%; POD3=79.0%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	6680	442	0
	4.2 < ws < 7.6	492	5837	380
	ws > 7.6	0	371	5664
Number of points 19866 ; PC=91.5%; EBD=0.0%; POD1=93.1%; POD2=87.8%; POD3=93.7%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed		Observations		
ms^{-1}		ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	17858	155	0
	12.0 < ws < 18.0	123	1396	39
	ws > 18.0	0	61	234
Number of points 19866 ; PC=98.1%; EBD=0.0%; POD1=99.3%; POD2=86.6%; POD3=85.7%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	7927	460	3		111
	East	749	1420	337		10
	South	71	430	2886		121
	West	300	91	12		213
Number of points 15141 ; PC=82.2%; EBD=1.2%; POD1=87.6%; POD2=59.1%; POD3=89.1%; POD4=46.8%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	8887	119	0		31
	East	136	2192	85		0
	South	0	90	3145		12
	West	24	0	8		412
Number of points 15141 ; PC=96.7%; EBD=0.0%; POD1=98.2%; POD2=91.3%; POD3=97.1%; POD4=90.5%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3763	1264	9
	7.1 < rh < 15.3	2514	3550	711
	rh > 15.3	300	2076	5403
Number of points 19590 ; PC=64.9%; EBD=1.6%; POD1=57.2%; POD2=51.5%; POD3=88.2%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	6222	296	3
	7.1 < rh < 15.3	355	6364	136
	rh > 15.3	0	230	5984
Number of points 19590 ; PC=94.8%; EBD=0.0%; POD1=94.6%; POD2=92.4%; POD3=97.7%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	18988	44	0
	50.0 < rh < 70.0	48	473	6
	rh > 70.0	0	0	31
Number of points 19590 ; PC=99.5%; EBD=0.0%; POD1=99.7%; POD2=91.5%; POD3=83.8%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm	Observations			
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6606	1298	10
	1.9 < pwv < 3.5	1105	3463	637
	pwv > 3.5	62	734	5515
Number of points 19430 ; PC=80.2%; EBD=0.4%; POD1=85.0%; POD2=63.0%; POD3=89.5%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm	Observations			
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	7636	182	0
	1.9 < pwv < 3.5	137	5245	81
	pwv > 3.5	0	68	6081
Number of points 19430 ; PC=97.6%; EBD=0.0%; POD1=98.2%; POD2=95.5%; POD3=98.7%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	2029	2037	679
Model data 0.6 < see < 0.8	1397	1988	1765
see > 0.8	161	593	1650
Number of points 12299 ; PC=46.1%; EBD=6.8%; POD1=56.6%; POD2=43.0%; POD3=40.3%			

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	3590	759	30
Model data 0.6 < see < 0.8	542	3400	418
see > 0.8	6	514	3565
Number of points 12299 ; PC=82.3%; EBD=0.3%; POD1=86.8%; POD2=72.8%; POD3=88.8%			

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	3460	0	431
Model data 0.6 < see < 0.8	0	4504	431
see > 0.8	127	114	3663
Number of points 12299 ; PC=94.5%; EBD=4.5%; POD1=96.5%; POD2=97.5%; POD3=81.0%			

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	4132	24	28
Model data 0.6 < see < 0.8	0	4630	27
see > 0.8	6	19	3985
Number of points 12299 ; PC=99.4%; EBD=0.3%; POD1=99.9%; POD2=99.1%; POD3=98.6%			

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1197	774	57
	$3.4 < \tau < 5.8$	647	2272	1170
	$\tau > 5.8$	64	1191	3553
Number of points 10925 ; PC=64.3%; EBD=1.1%; POD1=62.7%; POD2=53.6%; POD3=74.3%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1750	224	3
	$3.4 < \tau < 5.8$	158	3736	346
	$\tau > 5.8$	0	277	4431
Number of points 10925 ; PC=90.8%; EBD=0.0%; POD1=91.7%; POD2=88.2%; POD3=92.7%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1697	135	469
	$3.4 < \tau < 5.8$	147	3407	469
	$\tau > 5.8$	64	695	4311
Number of points 10925 ; PC=86.2%; EBD=4.9%; POD1=88.9%; POD2=80.4%; POD3=82.1%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1906	2	32
	$3.4 < \tau < 5.8$	2	4198	32
	$\tau > 5.8$	0	37	4748
Number of points 10925 ; PC=99.3%; EBD=0.3%; POD1=99.9%; POD2=99.1%; POD3=98.7%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2003	2188
	0.6 < glf < 0.7	1090	1408
	glf > 0.7	143	543
Number of points 11048 ; PC=37.4%; EBD=13.7%; POD1=61.9%; POD2=34.0%; POD3=19.7%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2742	633
	0.6 < glf < 0.7	470	2964
	glf > 0.7	24	542
Number of points 11048 ; PC=78.6%; EBD=0.4%; POD1=84.7%; POD2=71.6%; POD3=81.2%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3065	620
	0.6 < glf < 0.7	28	3501
	glf > 0.7	143	18
Number of points 11048 ; PC=79.3%; EBD=14.7%; POD1=94.7%; POD2=84.6%; POD3=45.2%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3202	41
	0.6 < glf < 0.7	10	4088
	glf > 0.7	24	10
Number of points 11048 ; PC=99.0%; EBD=0.5%; POD1=98.9%; POD2=98.8%; POD3=98.6%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3414	861
	4.6 < ws < 7.4	2929	4211
	ws > 7.4	271	2081
Number of points 19948 ; PC=62.4%; EBD=2.2%; POD1=51.6%; POD2=58.9%; POD3=78.1%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	5869	473
	4.6 < ws < 7.4	742	6294
	ws > 7.4	3	386
Number of points 19948 ; PC=90.3%; EBD=0.0%; POD1=88.7%; POD2=88.0%; POD3=94.6%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	17441	190
	12.0 < ws < 18.0	129	1634
	ws > 18.0	0	64
Number of points 19948 ; PC=97.7%; EBD=0.0%; POD1=99.3%; POD2=86.5%; POD3=86.3%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations West
Model data	North	6071	512	2	163
	East	134	416	84	0
	South	33	21	1516	458
	West	2294	29	206	5279
Number of points 17218 ; PC=77.1%; EBD=0.4%; POD1=71.2%; POD2=42.5%; POD3=83.8%; POD4=89.5%					

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations West
Model data	North	8195	70	2	201
	East	67	901	10	0
	South	0	5	1711	76
	West	270	2	85	5623
Number of points 17218 ; PC=95.4%; EBD=0.0%; POD1=96.1%; POD2=92.1%; POD3=94.6%; POD4=95.3%					

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4554	1110	8
	8.9 < rh < 17.9	2385	4409	1060
	rh > 17.9	166	1566	4636
Number of points 19894 ; PC=68.4%; EBD=0.9%; POD1=64.1%; POD2=62.2%; POD3=81.3%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	6766	320	0
	8.9 < rh < 17.9	339	6575	143
	rh > 17.9	0	190	5561
Number of points 19894 ; PC=95.0%; EBD=0.0%; POD1=95.2%; POD2=92.8%; POD3=97.5%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	19203	34	0
	50.0 < rh < 70.0	28	613	7
	rh > 70.0	0	2	7
Number of points 19894 ; PC=99.6%; EBD=0.0%; POD1=99.9%; POD2=94.5%; POD3=50.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6660	1065	0
	2.1 < pwv < 3.7	1350	3544	711
	pwv > 3.7	87	930	5259
Number of points 19606 ; PC=78.9%; EBD=0.4%; POD1=82.3%; POD2=64.0%; POD3=88.1%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	7909	93	0
	2.1 < pwv < 3.7	188	5347	80
	pwv > 3.7	0	99	5890
Number of points 19606 ; PC=97.7%; EBD=0.0%; POD1=97.7%; POD2=96.5%; POD3=98.7%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4487	1798	191
	4.2 < ws < 7.6	2574	3371	1179
	ws > 7.6	400	1604	4697
Number of points 20301 ; PC=61.8%; EBD=2.9%; POD1=60.1%; POD2=49.8%; POD3=77.4%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	8049	468	33
	East	682	1494	363
	South	90	382	2840
	West	316	86	71

Number of points 15371 ; PC=82.2%; EBD=1.4%; POD1=88.1%; POD2=61.5%; POD3=85.9%; POD4=50.3%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3727	1280
	7.1 < rh < 15.3	2693	3594
	rh > 15.3	392	2182

Number of points 19995 ; PC=63.4%; EBD=2.2%; POD1=54.7%; POD2=50.9%; POD3=87.4%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6566	1539
	1.9 < pwv < 3.5	1292	3322
	pwv > 3.5	84	813

Number of points 19840 ; PC=76.2%; EBD=0.6%; POD1=82.7%; POD2=58.5%; POD3=84.2%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2340	2300	868
	0.6 < see < 0.8	1528	2289	1931
	see > 0.8	227	758	2332
Number of points 14573 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3923	0	594
	0.6 < see < 0.8	0	5167	594
	see > 0.8	172	180	4537
Number of points 14573 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2074	1075	147
	$3.4 < \tau < 5.8$	728	2482	1255
	$\tau > 5.8$	84	1384	3798
Number of points 13027 ; PC=64.1%; EBD=1.8%; POD1=71.9%; POD2=50.2%; POD3=73.0%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2610	200	591
	$3.4 < \tau < 5.8$	192	3982	591
	$\tau > 5.8$	84	759	4609
Number of points 13027 ; PC=86.0%; EBD=5.2%; POD1=90.4%; POD2=80.6%; POD3=79.6%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2268	2515
	0.6 < glf < 0.7	1437	1942
	glf > 0.7	247	530
Number of points 13216 ; PC=38.1%; EBD=13.9%; POD1=57.4%; POD2=38.9%; POD3=19.3%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3654	776
	0.6 < glf < 0.7	51	4169
	glf > 0.7	247	42
Number of points 13216 ; PC=78.5%; EBD=14.9%; POD1=92.5%; POD2=83.6%; POD3=45.0%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3429	877	201
	4.6 < ws < 7.4	3111	4119	1223
	ws > 7.4	282	2280	4779
Number of points 20301 ; PC=60.7%; EBD=2.4%; POD1=50.3%; POD2=56.6%; POD3=77.0%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	6092	583	13
	East	102	326	68
	South	36	49	1508
	West	2304	20	299

Number of points 17426 ; PC=76.4%; EBD=0.4%; POD1=71.4%; POD2=33.3%; POD3=79.9%; POD4=89.4%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4232	1043
	8.9 < rh < 17.9	2639	4203
	rh > 17.9	234	1839

Number of points 20229 ; PC=65.7%; EBD=1.3%; POD1=59.6%; POD2=59.3%; POD3=80.3%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6423	1197
	2.1 < pwv < 3.7	1579	3367
	pwv > 3.7	100	989

Number of points 19941 ; PC=75.9%; EBD=0.6%; POD1=79.3%; POD2=60.6%; POD3=85.1%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3806	1843	226
	4.2 < ws < 7.6	3055	3177	1309
	ws > 7.6	472	1756	4523
Number of points 20167 ; PC=57.1%; EBD=3.5%; POD1=51.9%; POD2=46.9%; POD3=74.7%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	7780	522	46
	East	917	1414	466
	South	189	435	2688
	West	289	60	47

Number of points 15351 ; PC=78.5%; EBD=1.9%; POD1=84.8%; POD2=58.2%; POD3=82.8%; POD4=34.5%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3449	1226
	7.1 < rh < 15.3	2829	3534
	rh > 15.3	496	2240

Number of points 19861 ; PC=61.1%; EBD=3.2%; POD1=50.9%; POD2=50.5%; POD3=84.6%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6562	1687
	1.9 < pwv < 3.5	1057	2942
	pwv > 3.5	223	979

Number of points 19674 ; PC=72.9%; EBD=1.7%; POD1=83.7%; POD2=52.5%; POD3=77.9%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2261	2153	859
	0.6 < see < 0.8	1471	2213	1813
	see > 0.8	332	956	2405
Number of points 14463 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3830	0	568
	0.6 < see < 0.8	0	5169	568
	see > 0.8	234	153	4509
Number of points 14463 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2010	1074	153
	$3.4 < \tau < 5.8$	762	2353	1164
	$\tau > 5.8$	111	1483	3813
Number of points 12923 ; PC=63.3%; EBD=2.0%; POD1=69.7%; POD2=47.9%; POD3=74.3%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2497	168	560
	$3.4 < \tau < 5.8$	275	3883	560
	$\tau > 5.8$	111	859	4570
Number of points 12923 ; PC=84.7%; EBD=5.2%; POD1=86.6%; POD2=79.1%; POD3=80.3%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2030	2280
	0.6 < glf < 0.7	1589	2158
	glf > 0.7	322	507
Number of points 13112 ; PC=39.2%; EBD=13.7%; POD1=51.5%; POD2=43.6%; POD3=22.6%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3558	695
	0.6 < glf < 0.7	61	4219
	glf > 0.7	322	31
Number of points 13112 ; PC=79.4%; EBD=14.6%; POD1=90.3%; POD2=85.3%; POD3=48.5%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3343	892	205
	4.6 < ws < 7.4	3160	4223	1240
	ws > 7.4	312	2168	4758
Number of points 20301 ; PC=60.7%; EBD=2.5%; POD1=49.1%; POD2=58.0%; POD3=76.7%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5881	597	20
	East	128	316	59
	South	90	38	1455
	West	2456	34	353

Number of points 17442 ; PC=73.9%; EBD=0.8%; POD1=68.7%; POD2=32.1%; POD3=77.1%; POD4=87.0%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3947	966
	8.9 < rh < 17.9	2813	4144
	rh > 17.9	345	1975

Number of points 20229 ; PC=63.3%; EBD=1.8%; POD1=55.6%; POD2=58.5%; POD3=77.9%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6199	1355
	2.1 < pwv < 3.7	1783	3132
	pwv > 3.7	120	1066

Number of points 19929 ; PC=72.8%; EBD=0.8%; POD1=76.5%; POD2=56.4%; POD3=82.6%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration